

CLUB club

2026-09-25

2026-09-24 class

Last week we learned

- How to build a bowtie2 "index".
- How to "map" FASTQ files against that index.
- About the contents of the resulting SAM file.

Today we'll learn how to do that at scale on a `Slurm` cluster.

But first

Public service announcement

■ Transferring files to/from the cluster

Till figured out that on the cluster you can access the Charité S drive as follows:

```
$ cd /charite-store-f/f-cc11-virologie/
```

```
(workspace) [jonestc@sc-frontend1 f-cc11-virologie]$ ls -l
total 1056
drwxrwx---  8 4294967292 wheel      230 Mar 28 15:11 Admin
drwxrwx---+ 20 bobo10      wheel      756 Sep 21 17:51 AG_Bogdanow
drwxrwx---+  3 root        wheel       27 Mar 23  2026 AG_Bogner
drwxrwx--- 13 srv-radenkom wheel     582 Sep 23 11:33 AG_Corman
drwxrwx---  4 root        wheel       88 Apr 29 10:54 AG_Drexler
drwxrwx--- 38 srv-radenkom wheel    1237 Sep  3 10:54 AG_Drosten
drwxrwx---+  7 smolkan     wheel     204 Jul 30 14:50 AG_Ghana_Team
drwxrwx---+ 18 nobody      wheel     541 Jun  7  2021 AG_Hanta
drwxrwx---+ 15 srv-radenkom wheel     574 Sep 24 20:49 AG_Jones
drwxrwx--- 10 srv-radenkom wheel     436 Sep 17 13:56 AG_Junglen
drwxrwx--- 22 nobody      wheel    1093 Sep 23 13:35 AG_Mueller
drwxrwx---+ 15 srv-radenkom wheel     546 Jul  3  2023 AG_Muth
drwxrwx--- 20 srv-radenkom wheel     559 Mar 28 11:44 AG_Niemeyer
drwxrwx--- 51 4294967292 wheel    28970 Jan 25  2021 AG_Reuter
drwxrwx---  5 root        wheel       95 Mar 30 11:23 AG_Schoenrich
drwxrwx---  3 4294967292 wheel       23 Mar 28 11:08 AG_Tenner
drwxrwx---+  7 srv-radenkom wheel     290 May  8 10:34 AG_Weiss
drwxrwx---+  5 srv-radenkom wheel     133 Mar 23  2026 Alt-Daten
drwxrwx---+  2 root        wheel        0 Mar 16  2026 Bestellungen
drwxrwx---+ 30 srv-radenkom wheel    1183 Sep 23 10:47 Daten
drwxrwx---+  6 walperf     posix-nogroup 163 Jul 17 10:33 Datentransfer
drwxrwx---+  7 semmleri     posix-nogroup 472 Aug 20 15:07 'DZIF-TTU EI-PM'
d---rwx---  8 srv-mmatthia wheel     741 Sep 15 12:37 Intern
drwxrwx--- 15 srv-radenkom wheel     700 Sep 24 14:50 'Klinische Virologie'
drwxrwx---  7 4294967292 wheel     133 Aug 19 13:50 Zentral
```

Slurm

The Simple Linux Utility for Resource Management.

1. Allocates access to compute "nodes" to users for some duration.
2. Provides a framework for starting, executing, and monitoring work on allocated nodes.
3. Arbitrates contention for resources by managing a queue of pending jobs.

Adapted from https://en.wikipedia.org/wiki/Slurm_Workload_Manager

High-performance compute system (HPCS)

Also known as a "cluster", an HPCS is a collection of (often diverse) computer "nodes" and with shared disk space, connected by a fast network, and centrally managed.

■ login and compute nodes

The nodes are of two types, `login` and `compute`:

- When you connect via `ssh` you will be on a login node. You're not supposed to do (and often are prevented from doing) much on these nodes. You use login nodes to obtain access to the compute nodes.
- Compute nodes typically have many CPUs and large amounts of RAM. The vocabulary around compute nodes is unfortunately (necessarily) a bit complicated. Today we will try to keep things simple.

When you connect via a Jupyter, you are assigned a compute node.

■ A scheduler

There will typically be many other users trying to use the HPCS

So, if you log in to a busy cluster, it will often be the case that all the nodes in the cluster are already running jobs for other people. And there may be many jobs already queued up waiting to run.

The "scheduler" (a program) takes care of managing resources.

The Slurm scheduler is very complicated!

Today

We'll learn something about using Slurm.

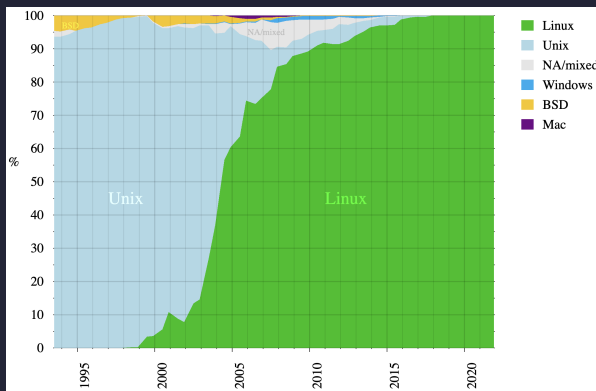
This is a core reason for you attending this course.

Slurm is running on the majority of the top 500 HPCS systems.

So what you learn here should be applicable wherever you wind up (at least if you're in academia). The BIH cluster also runs Slurm.

Linux runs on all of the top 500 HPCS systems.

Note that there are scheduling systems other than Slurm and there are also powerful single computers that can be accessed with no need to use a scheduler. In a future class we could also learn how you can access cloud compute systems (e.g., Amazon EC2).



A little about the Charité cluster

You can use the `sinfo` command to get a summary of the nodes in the cluster.

```
$ sinfo -o "%P %N %c %m"
PARTITION  NODELIST                                CPUS  MEMORY
compute*   s-sc-node[002-044]                      64+    504811+
gpu         s-sc-gpu[002-029]                       64     504811
pgpu        s-sc-dgx01,s-sc-pgpu[01-08,11-16]    64+    504811+
agpu        s-sc-agpu001                        64     1030000
```

Note that I added some whitespace padding to the output above.

Use `sinfo -N -o "%P %N %c %m" | less` if you want to see a line of output for each node.

Nodes are divided into "partitions".

See `man sinfo` (and other manual pages!) for more detail.

And <https://slurm.schedmd.com/documentation.html> is very useful, as is Claude.

Getting compute nodes

You will typically want to do one of two things:

- Get a shell so you can interactively run a series of commands on a node with lots of memory and/or CPUs.
- Run a certain command (or commands) that you know will require lots of memory and/or CPUs.

These two have much in common.

The main difference is that in the first, you wait for the Slurm allocation, then type commands. Every command you type will be run on the node with all the resources available. You don't need to know in advance exactly what you're going to want/need to do.

In the second, you already know the commands you want to run and you submit them to Slurm and it takes care of running them.

You can use the first approach to do the second, but if you do it the first way you will need to sit around waiting for the compute node to become available. That could take hours or even days, depending on what you're requesting.

We'll look at both.

Let's look around a bit

First of all, let's look around a bit. You can use `nproc` to see how many "cores" are available to you (a "thread" of execution can run on each core). Also look at the environment variable `SLURM_MEM_PER_NODE` and at your host name.

After I log in, I see this:

```
$ nproc
16

# This doesn't show anything as I'm not on a compute node so the
# variable isn't set.
$ echo $SLURM_MEM_PER_NODE

$ hostname
s-sc-frontend1
```

You will see something different if you do this in Jupyter because you're already on a compute node.

Submit a job with sbatch

Put this in a file called `date.sh`:

```
#!/bin/bash

#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --time=2:00

sleep 30
date
```

Then hand it to the Slurm scheduler using the `sbatch` command

```
$ sbatch date.sh
Submitted batch job 10906138

$ squeue -u $USER
JOBID    PARTITION   NAME     USER      ST  TIME  NODES   NODELIST(REASON)
10906138 compute    date.sh  jonestc    R   0:01     1     s-sc-node004

$ ls -l
-rw-r--r-- 1 jonestc posix-nogroup 110 Sep 24 22:00 date.sh
-rw-r--r-- 1 jonestc posix-nogroup  33 Sep 24 21:59 slurm-10906138.out
```


A necessary evil

(And pain in the butt)

You need to estimate in advance how much memory your job will need, how many cores will be a good number, and how long the job will take.

It can be hard to do this, and sometimes frustrating!

Get a node to run `bash`, but just for 10 seconds:

```
srun --pty --mem 4G --cpus-per-task 8 --time 00:00:10 bash -i
```

Shortly afterwards...

```
[2026-09-25T01:48:04.004] error: *** STEP 10907405.0 ON s-sc-node008 CANCELLED AT  
2026-09-25T01:48:04 DUE TO TIME LIMIT ***  
srun: Job step aborted: Waiting up to 32 seconds for job step to finish.
```

Note: some long-running commands can save a periodic status file. This allows them to be restarted if your job exceeds the amount of time you allocated for it.

Slurm terminology - keeping it as simple as possible

If you just want to run one process on one node and only specify the memory and number of cores, you don't need to learn many terms or command-line arguments.

Job = the allocation

```
sbatch --nodes=1 --ntasks=1 --cpus-per-task=32 --mem=64G
```

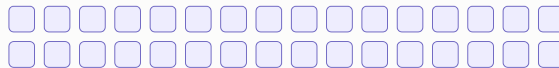
Node

Memory limit: 64G (--mem)

Batch step: the script runs bowtie2 directly

One task: the bowtie2 process

bowtie2 -p 32 runs 32 threads

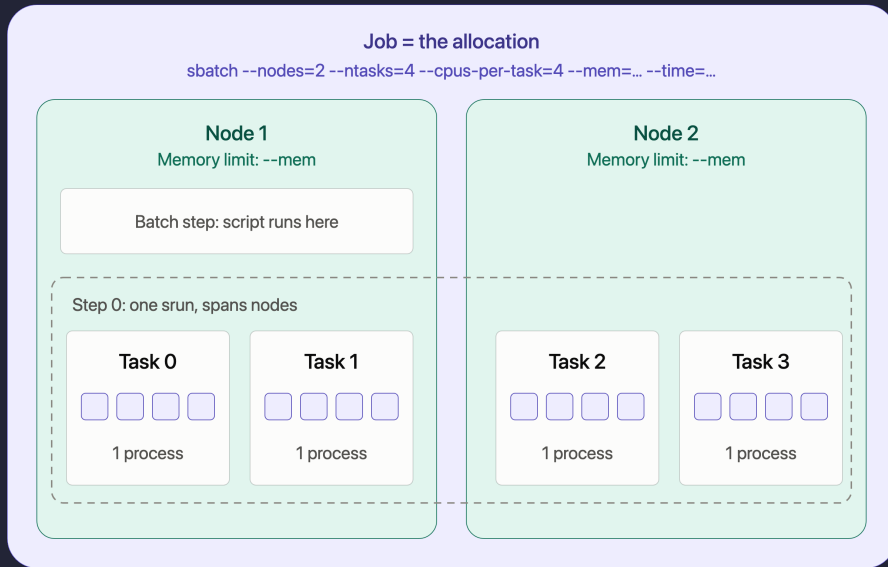


32 CPUs (--cpus-per-task=32), one per thread

■ = 1 CPU. All threads share the task's CPUs and the node's 64G.

Slurm terminology - more detail

If you want to do something more complex, the terminology can be quite confusing. Claude can explain it pretty well, though, if you keep on asking questions and for clarifications.



■ = 1 CPU. Squares per task: `--cpus-per-task`. Number of tasks: `--ntasks`.

Dashed box = step. Each srun creates a new step; its tasks share each node's memory limit.

Let's do some real work

Here's how you could run the `bowtie2` command from last week using Slurm, requesting 16 Gb of RAM and 32 cores on a node for one hour.

```
$ cat bowtie2-submit.sh
#!/bin/bash

#SBATCH --cpus-per-task=32
#SBATCH --time=1:00:00
#SBATCH --mem=16G

# You can get into a pixi shell like this (for all commands)
# eval "$(pixi shell-hook --manifest-path ~/workspace/pixi.toml --shell bash)"

# Or just use 'pixi run' before your commands.
pixi run bowtie2 \
  --threads $SLURM_CPUS_PER_TASK \
  --local --xeq --no-unal -x ../20260918-map/HBV \
  -1 ~/data/fastq/HBV_RISE718_R1_trimmed.fastq.gz \
  -2 ~/data/fastq/HBV_RISE718_R2_trimmed.fastq.gz \
  -U ~/data/fastq/HBV_RISE718_single_trimmed.fastq.gz \
  > matches-local-with-single.sam

# Then give it to sbatch to schedule the job:
$ sbatch bowtie2-submit.sh
Submitted batch job 10907434
```

Note that my `bowtie2` HBV database is in `../20260918-map/HBV`. Yours will almost certainly be somewhere else, so you'll need to adjust the `-x` argument in the above.

Even more fun

And here's a run on the 0.5 billion reads:

```
$ cat bowtie2-big-submit.sh
#!/bin/bash

#SBATCH --cpus-per-task=64
#SBATCH --time=12:00:00
#SBATCH --mem=128G

pixi run bowtie2 \
  --threads $SLURM_CPUS_PER_TASK \
  --local --xeq --no-unal -x ../20260918-map/HBV \
  -1 ~/data/fastq/LV7008876184-LV7008414499-CGG2015787_S1_L002_R1_001-ar3.fastq.gz \
  -2 ~/data/fastq/LV7008876184-LV7008414499-CGG2015787_S1_L002_R2_001-ar3.fastq.gz \
  > matches-local-big-FASTQ.sam

# Then give it to sbatch to schedule the job:
$ sbatch bowtie2-submit.sh
Submitted batch job 10906811
```